

# Cortical cell and neuron density estimates in one chimpanzee hemisphere

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**The density of cells and neurons in the neocortex of many mammals varies across cortical areas and regions. This variability is, perhaps, most pronounced in primates. Nonuniformity in the composition of cortex suggests regions of the cortex have different specializations. Specifically, regions with densely packed neurons contain smaller neurons that are activated by relatively few inputs, thereby preserving information, whereas regions that are less densely packed have larger neurons that have more integrative functions. Here we present the numbers of cells and neurons for 742 discrete locations across the neocortex in a chimpanzee. Using isotropic fractionation and flow fractionation methods for cell and neuron counts, we estimate that neocortex of one hemisphere contains 9.5 billion cells and 3.7 billion neurons. Primary visual cortex occupies 35 cm<sup>2</sup> of surface, 10% of the total, and contains 737 million densely packed neurons, 20% of the total neurons contained within the hemisphere. Other areas of high neuron packing include secondary visual areas, somatosensory cortex, and prefrontal granular cortex. Areas of low levels of neuron packing density include motor and premotor cortex. These values reflect those obtained from more limited samples of cortex in humans and other primates.**

primate neocortex | neuron density | visual cortex | isotropic fractionator | flow fractionator

**T**he present study is part of our ongoing effort to determine how human brains are both similar and different from the brains of our closest living primate relatives (e.g., refs. 1 and 2). Present-day chimpanzees and bonobos diverged from the line of hominins that led to modern humans some 6 million years ago (3). The modern chimpanzee brain is similar in size to our earliest hominin ancestors (4). In contrast, the brains of later hominins rapidly increased in size about 1.5 million years ago, resulting in modern humans having brains about three times the size of early hominins and present-day chimpanzees. Similarly, the neocortical sheet of one cerebral hemisphere is almost three times larger in humans (average, ~975 cm<sup>2</sup> in humans; e.g., refs. 5 and 6) than in chimpanzees (~341 cm<sup>2</sup>). This increase also appears to be reflected in an increase in the number of areas that are architectonically and physiologically modified for different specialized functions. Such specializations are reflected in a wide range of neuron types and laminar and sublaminal patterns of regional and areal organization in larger primate brains, especially those of humans (7). It would be surprising if average neuron size and neuron packing densities did not vary across areas, especially in large-brained primates. However, the claim has been made that neuron densities are uniform across cortical areas and mammalian species, except for a twofold increase in primary visual cortex (V1) in some primates (8, 9). This conclusion is contrary to the results of several previous studies (e.g., ref. 10), especially those of our recent reports on neuron densities across the cortical sheet in several primate species (1, 2). The primates examined to date (baboons, macaques, and galagos) had much higher neuron densities in V1. Neuron densities were also higher than average in other sensory areas, and

considerably lower in primary motor cortex (M1), with more pronounced differences appearing in larger-brained primates. As there is a relationship between neuron density in cortex and average neuron size (1, 11), and larger neurons have larger dendritic arbors with more synaptic contacts (12), larger neurons are more suitable for integrative functions, whereas smaller neurons preserve information and more faithfully reflect the sources of their activation (13–15).

Here, we extend our observations on areal neuron densities in strepsirrhine galagos and anthropoid monkeys to the full neocortical sheet of a chimpanzee (*Pan troglodytes*). We provide the most detailed assessment of cortical neuron densities across the cortical sheet for any species, having dissected the chimpanzee cortex into 742 tissue pieces, providing roughly six times more detail than has ever been reported for the cortex of any primate species (1) and nearly 100 times more evaluations than a typical quantitative study (10). This comprehensive approach allows us to demonstrate significant regional differences in neuron densities that may be small enough to be lost in the noise of more limited studies. Although this has not yet been attempted for a human brain, direct comparisons will be possible in future studies.

## Results

The total surface area of the cerebral cortex of the right hemisphere of this chimpanzee was 341 cm<sup>2</sup>, although that could be a slight underestimate resulting from a loss of some neocortex along the outer margins of the cortical sheet. The neocortex contained an estimated 9.51 billion cells (count includes all cell types identified by DAPI; e.g., glial, neuronal, etc.), of which about 3.71 billion, or 39%, were neurons (only cells that stain positive for neuronal nuclear antigen). The cell and neuron

## Significance

**Chimpanzees are our closest relatives, and understanding the organization of their brains can help us understand our own evolution. Here we present a detailed examination of cell and neuron densities across the chimpanzee cortex. We found similarities to other mammals, including primary sensory areas with high neuron densities and a trend of decreasing neuron densities along the posterior to anterior axis of the cortex. However, we also found a prefrontal region with anomalously high neuron density that disrupts the trend of decreased neuron densities in frontal brain regions. The data reported here allow valuable comparisons among the brains of our close relative and those of humans and other primates.**

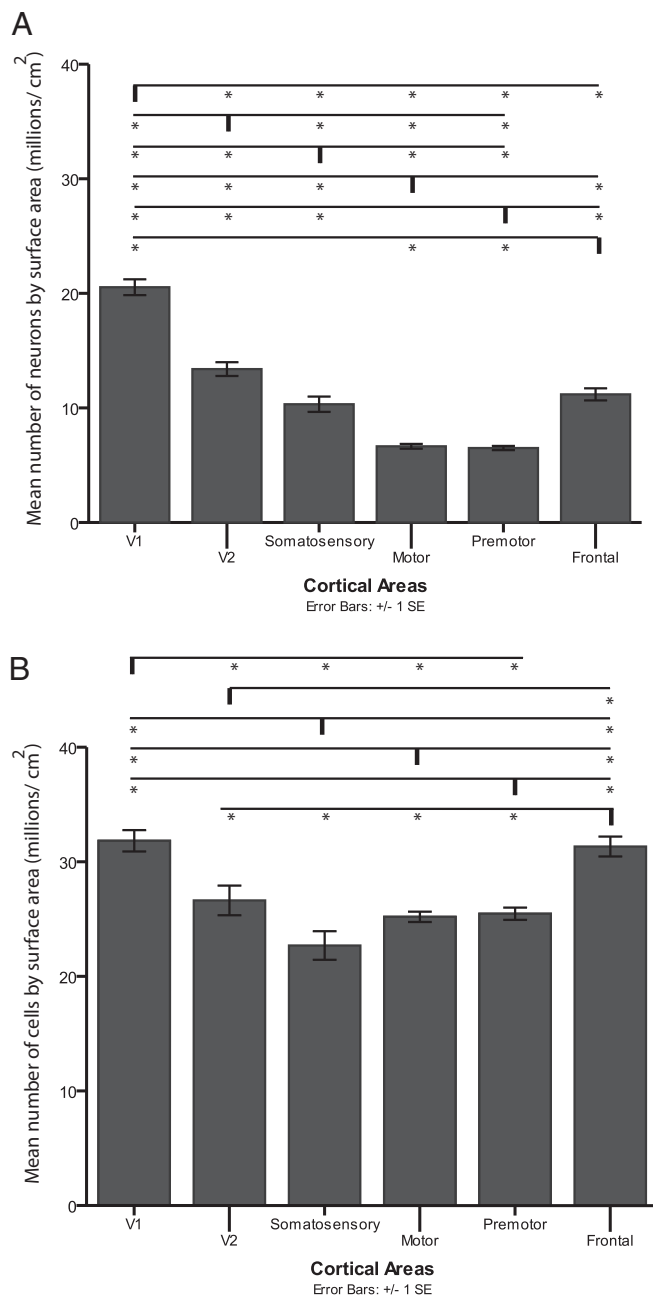
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**Fig. 2.** Pairwise comparison results of estimated marginal means based on the original scale of the dependent variable, either neuron (A) or cell (B) density in square centimeter, with mean differences significant at  $P < 0.05$  and robust SE bars shown. All significant pairwise comparisons are highlighted by the significance bars. (A) V1 estimated marginal means of neuron density are significantly higher than all other cortical regions, whereas V2 is significantly higher than motor, premotor, and somatosensory blocks. Somatosensory predicted values are also significantly higher than motor and premotor blocks. Motor and premotor cortices do not significantly differ from one another, but are both significantly lower than all other areas of cortex. (B) V1 and the frontal region estimates show no significant differences in cell density, but each region independently contains higher means than every other area. Somatosensory, premotor, and motor cortex do not significantly differ from one another.

Neuron numbers averaged 10 million per square centimeter (Fig. 2A), or 41 million neurons per gram of tissue. The somatosensory section of cortex contained 362 million cells, of which 164 million, or 45%, were neurons.

**Motor and Premotor Areas.** Our block of tissue pieces designated as motor in Fig. 1 likely contains most of M1, including the cortex of the rostral bank of the central sulcus and cortex of the caudal half of the precentral gyrus (17, 23–25). The more rostral block of pieces likely includes dorsal and ventral divisions of premotor cortex. Our motor cortex block consisted of 24.97 cm<sup>2</sup> of cortical surface with 625 million cells, of which 163 million, or 27%, were neurons. Cell packing densities averaged 25 million cells per square centimeter of cortical surface (Fig. 2B), or 91 million cells per gram of tissue. Neuron packing densities averaged 7 million neurons per square centimeter of surface (Fig. 2A), or 24 million neurons per gram. Nearly all of the individual pieces of M1 had low neuron densities, in the 6–7 million range. Thus, M1 was characterized by low neuron packing densities and moderate cell packing densities.

Our premotor block included 24.46 cm<sup>2</sup> of cortical surface, with 636 million cells and 172 million neurons, for a composition of 26% neurons. Cell densities in premotor cortex averaged 25 million per square centimeter of cortical surface (Fig. 2*B*), or 88 million per gram of tissue. The average neuron packing density was 6 million neurons per square centimeter of cortical surface (Fig. 2*A*), or 23 million neurons per gram of cortical tissue. Thus, cortex in the premotor block had low levels of neuron packing, with little variability across tissue pieces, and closely matched the low neuron packing densities across motor areas.

**Prefrontal Cortex.** Prefrontal cortex generally consists of the cortex rostral to premotor cortex, and it has several subdivisions including a large dorsolateral region of granular frontal cortex and adjoining regions of orbital frontal and medial frontal cortex (26–28). Areal variations in neuron densities across prefrontal cortex have previously been shown in macaques (29). Overall, the prefrontal block of tissue denoted in Fig. 1 *B* and *C* has pieces of tissue with higher cell packing densities and neuron packing densities than those in either motor or premotor cortex. Most of this cortex with higher cell and neuron packing densities would be considered granular frontal cortex. Pieces of cortex with lower values were located along the margins of frontal cortex, including medial frontal and orbital frontal regions. However, we distinguished a dorsomedial block of cortex within the presumptive region of granular frontal cortex (Fig. 1*C*) as having higher neuron densities than other prefrontal regions. Pieces of cortex in this block had average neuron packing densities of 11 million per square centimeter. Total cell densities averaged 30 million per square centimeter, or 106 million per gram, with 36% neurons. This high-density block of tissue covered 13.11 cm<sup>2</sup> with 515 cells and 181 million neurons.

**The Anterior to Posterior Pattern.** To examine whether the chimpanzee brain showed uniform cell and neuron packing densities across cortex (or a linear decrease in density beyond V1 from posterior to anterior), we examined densities in pieces of cortex across the anterior-to-posterior (A-P) dimension, using linear regression and curve estimation, testing multiple model functions. Recently, there have been a number of descriptions of neuron packing densities across the cortical sheet in monkeys and other mammals that revealed a trend from low packing densities to high neuron packing densities from anterior to posterior cortex (11, 30–33). When we assessed how cell and neuron densities varied in the selected cortical regions across the A-P dimension, using generalized linear modeling with robust estimators, most cortical areas showed significant differences in estimated means of neuron and cell density (Fig. 2*A* and *B*). In addition, the locations of samples within cortical regions were better predictors than A-P coordinates alone of cell density (cortical area,  $P = 6.4 \times 10^{-10}$ ; A-P,  $P = 0.818$ ) and of neuron density (cortical area,  $P = 1.0 \times 10^{-4}$ ; A-P,  $P = 0.835$ ).

When the density values for individual pieces were plotted by A-P location and color-coded for tissue block of origin, the average curve





V1, and very low values for anterior slices in frontal and prefrontal cortex. It is not clear yet from these results whether V1 and V2 differ, whether M1 and premotor cortex are specifically low in neuron number, or whether a granular region of prefrontal cortex with higher neuron densities exists. It is also uncertain whether primary somatosensory cortex is higher in primates in general, as expected from its well-developed layer of granular cells, or whether granular primary auditory cortex values are higher in any primate. As for the nonprimates that have been studied, areal differences in mean densities may be less pronounced, but they vary considerably across cortical areas with high densities in V1 and primary somatosensory cortex (S1) in mice (39) and rats (41). Others have reported higher neuron densities in S1 than M1 and V2 in rats (42), and higher values in visual areas and S1 than M1 in cats (10).

**Does Neuron Packing in Chimpanzee Cortex Reflect a Developmental Pattern of Cortical Neurogenesis?** In a series of publications, Finlay and colleagues have presented evidence that an anterior to posterior pattern of cortical development that is seen in primates and other mammals results in a matching gradient of neuron densities from low to high across the cortex (e.g., refs. 11 and 31–33). Our chimpanzee results roughly reflect such a pattern, with at least four exceptions. These exceptions include the sharp increase in neuron densities in V1 from V2 at the V1–V2 border, the very low densities in motor and premotor cortex, the increased neuron densities in anterior somatosensory cortex, and the higher than expected neuron densities in dorsofrontal cortex of “granular” frontal cortex. These exceptions do not argue against the developmental gradient having an important role in creating neuron density differences across the cortex, but the exceptions do indicate that other factors are also involved.

Such additional factors may include areal differences in neuron death during development (31). Direct evidence for this possibility comes from V1 of macaques after loss of visual inputs to the brain during fetal development. In such monkeys, parts of V1 fail to develop normally histologically (43, 44), and these abnormal parts had cell densities reduced by about 25% (43). In view of the developmental gradient theory of regional differences in neuron densities, it may just be a fortunate circumstance for primates that V1 evolved in caudal neocortex. Higher neuron densities in V1 function to precisely preserve visual information, whereas motor and premotor cortex, as well as parts of prefrontal cortex, evolved within more anterior cortex to contain extremely low neuron densities as a result of larger neurons. This makes neurons in these anterior regions better suited to integrate information from many sources of activation. Alternatively, more modular features of cortical development could have played a prominent role in the evolution of such striking differences in neuron densities across cortex.

**Functional Implications.** Differences across cortical areas in neuron packing densities imply there is an inverse relationship with average neuron size (45). Larger neurons take up more space and require more glial and other support cells that vary in size. Smaller neurons have smaller dendritic arbors and are connected by fewer inputs (46). Overall, small cortical neurons are better designed for preserving information from a small number of activating inputs, whereas large neurons are better suited for integrating information from a larger number of activating inputs (47, 48). Thus, V1 has densely packed small neurons (granular cells) in layer 4 that are activated by just a few neurons in the dorsal lateral geniculate nucleus, and they activate other neurons in V1 that are, with few exceptions, small pyramidal neurons with small apical arbors contacted by relatively few inputs (46, 48). The high neuron densities for V1 of primates have been postulated as a mechanism for preserving the high visual acuity of primates (49). Motor cortex is known for its large pyramidal cells and lack of small layer 4 granular cells, which promotes integration from more sources of information. Primary sensory areas generally have smaller neurons with smaller dendritic arbors, whereas higher-

order sensory areas have larger neurons with larger arbors. The neuron density values in cortex illuminate this hypothesis in detail by indicating average levels of information preservation and integration for areas across the cortical sheet for chimpanzee. However, neurons of quite different sizes may occur in the same area and play different functional roles. For example, V1 contains both small granular cells in layer 4 and large Meynert cells in layers 5 and 6. However, the average neuron size should suggest a dominant role for a region or area.

One of the important findings of the present study was that a dorsal part of granular prefrontal cortex in the chimpanzee had a region of higher neuron density. As the term “granular frontal cortex” implies, this cortex contains small neurons in layer 4. However, large pyramidal neurons have also been described in prefrontal cortex, suggesting such neurons receive and sum many inputs (12). Although all parts of granular prefrontal cortex would seem suitable for preserving information, the dorsomedial part seems more specialized for this function. Notably, granular frontal cortex is thought to be a specialization of the primate brain (26) that appears to be important in working memory (50). Our present results are consistent with the conclusion that granular frontal cortex is not uniform in function (28), and the region of particularly dense neuron packing in frontal cortex may be a specialization of primates that occurs to a lesser extent in Old World macaques (1) and baboons (40), is especially marked in chimpanzees, and is likely in humans.

## Materials and Methods

Experimental procedures were all approved by the Vanderbilt Institutional Animal Care and Use Committee. One adult female chimpanzee brain was obtained for this study from the Texas Biomedical Research Institute. The age of the chimpanzee was estimated to be 53 y. The animal was humanely killed because of myocarditis (heart failure). The neocortex is expected to have few age-related changes in a chimpanzee of this age (51, 52). Shortly before death, her body weight was 34.8 kg. The brain was flushed with 0.9% PBS, removed from the skull, and shipped overnight in the same solution. On arrival, the brain weight was 344 g. The brain was bisected into right and left hemispheres. The right neocortex was flattened and used in this report, and the left hemisphere was sectioned and histologically processed for other studies.

To determine neuron and cell number per unit of cortical surface area across all parts of the cortical surface, we first manually flattened the neocortex into a sheet. The unfixed cortex of the right hemisphere was separated from the underlying structures and most of the white matter. A cut was made in the depth of the central sulcus to separate the most anterior region, and another cut was made to separate primary visual cortex from the rest of the caudal cortex, creating three separate cortical pieces for flattening. This was done to preserve the integrity of the tissue during flattening. Sulci were carefully opened, and the cortical sheet was then unfolded under gentle pressure (Fig. 1A). Excess white matter was removed. After postfixing in 4% (wt/vol) paraformaldehyde for 2 wk, each sheet was cut into small pieces, ~5 mm<sup>2</sup> in surface area, resulting in 742 pieces that were photographed, weighed, numbered, and assigned to a cortical area when possible (Fig. 1B and C). Pieces were assigned to cortical areas based on their expected relation to sulcal landmarks and their myelin content, as primary sensory areas appear dark relative to the surrounding cortex when viewed on a light box because of their myelin-dense composition. The surface area of each piece of cortex was measured from the photograph, using ImageJ (NIH). Each piece of cortex was individually disassociated to a solution, where cell membranes were ruptured but cell nuclei remained intact, as previously described (34, 36). We determined the number of cells in each tissue piece, using the isotropic fractionator method (36). The flow fractionator method was used to estimate of the total number of neuron nuclei labeled with the anti-NeuN antibody and DAPI in each tissue piece, as previously described in detail (1, 2, 34, 35, 40).

IBM-SPSS software (version 22) was used to test for deviations from the normal distribution (Kolmogorov-Smirnov test), to test for nonlinearities (linear regression), and to compare cell and neuron densities between selected cortical areas (generalized linear modeling). Significance was considered for  $P < 0.05$ . Statistical analysis included using Huber-White robust SEs through the generalized linear modeling procedures specifying “robust” covariance matrix estimators (also called Huber-White or sandwich estimators) in SPSS software. Robust SEs are corrections to help account for violations of the assumption of independence between cell counts on tissue pieces from a single chimpanzee brain. The resulting  $P$  values can be used to

help describe differences within the one brain, but they do not directly allow inferences about the population of chimpanzees.

Cell and neuron numbers per square centimeter of cortical surface were obtained for each piece of cortex and color coded for cells (Fig. 1B) or neurons (Fig. 1C). In addition, each dissected piece of tissue was assigned coordinates along the A-P axis by generating centroid measures for each piece, using NIH ImageJ software. Cell and neuron numbers per square centimeter of surface area versus the A-P locations were plotted and used in linear regression curve-fitting estimations to assess whether cell and neuron densities were uniform or increased or decreased in the A-P dimension with linear, quadratic, and cubic regression models.

The results were tested for deviations from the normal distribution (Kolmogorov-Smirnov test), and generalized linear regression modeling was then used to compare cell and neuron densities between selected cortical areas. Selected cortical areas defined by experts were the components of one

fixed factor (see Fig. 1C for regional bin map), and A-P coordinates (x-value from the centroid calculation) were binned for use as the components of the second fixed factor. Model main effects were tested and evaluated for the relative contribution of each factor to the variance in mean density. Multiple comparisons were adjusted by the Bonferroni method, and robust estimators, as addressed earlier, were used in this analysis to account for violations of independence between the samples from a single chimpanzee.

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